

Planning and Scheduling (Resource Planning)

Construction schedules

Schedules can be either time oriented or resource oriented

In one of the earlier discussion, we have discussed about the concepts related to time-oriented scheduling.

Let us look into resource-oriented scheduling concepts

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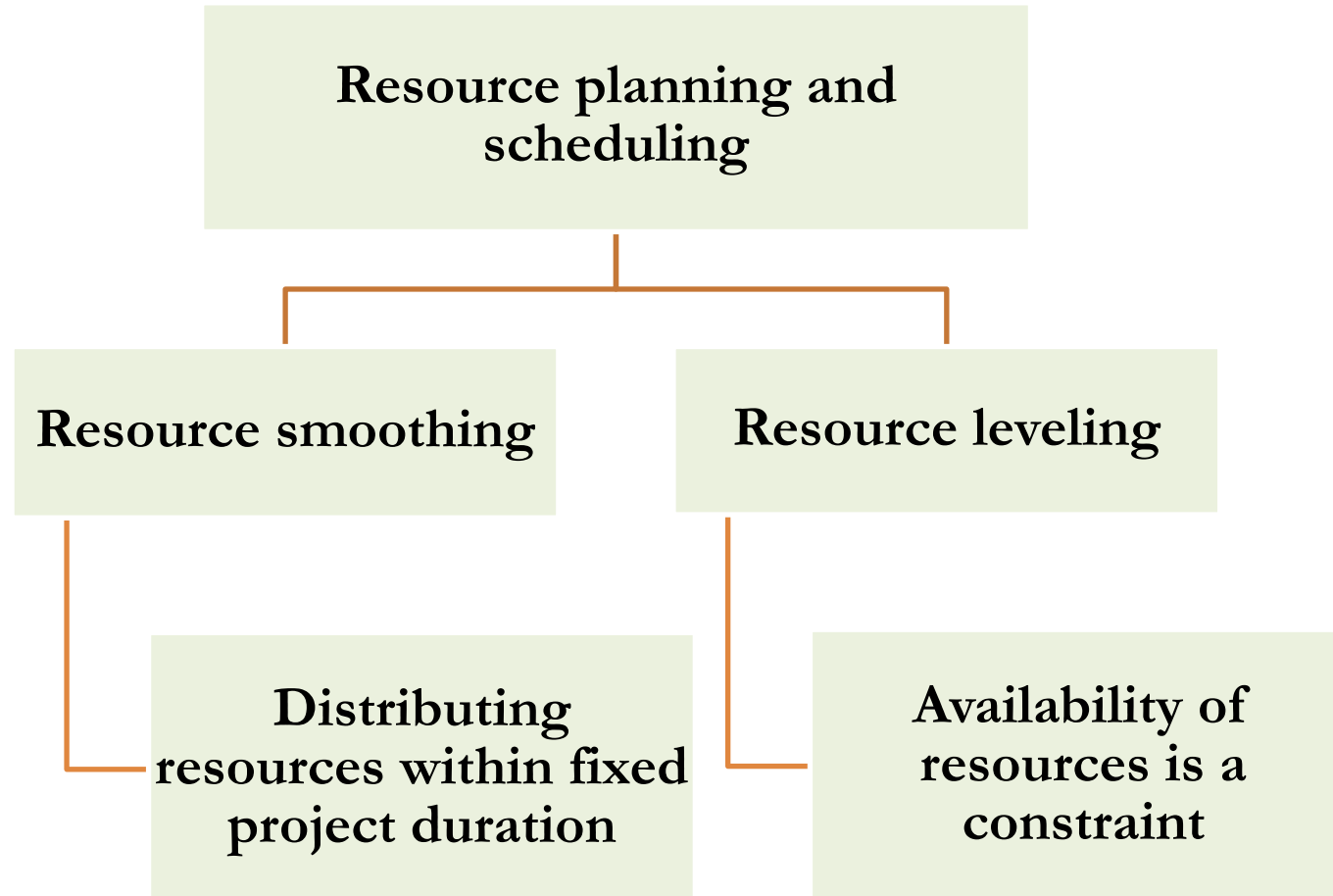
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Resource scheduling

- Estimation of activity times and the project duration using network diagrams assuming that there is an unlimited availability of resources.
- Real-life projects faces the resource constraints such as:-
 - Shortage of resources
 - Delayed delivery
 - Competing demands
 -
- Resource planning and allocation plays an important role in managing the construction projects.

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RESOURCE SMOOTHING

Resource smoothing is the process of allocating resources among various activities such that the peak requirement of resources is reduced with a constraint on time.

Underlying assumptions in resource smoothing are

- Only one type of resource is considered throughout the project
- Activities already underway should be completed first

Resource smoothing is carried out through the following steps:

1. Identification of critical activities
2. Arrange the activities on time axis based on their **early start dates**
3. The total number of resources required on each day are obtained by adding the resource requirements of individual activities on that particular day, and the resource profile determined as a chart called resource aggregation (or resource loading) chart
4. Repeat steps 2 and 3, with activities plotted using their **latest start dates**
5. Compare the two resource charts in steps 3 and 4
6. An acceptable resource profile between the two extremes is found by eliminating the peaks and valleys of the resource distributions. This is done by arranging the non-critical activities within their float, which obviously ensures that the duration of the project is not extended

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Consider a project having seven activities A, B, ... G. Duration (in days) of each activity and the daily resource requirement are given in the Table below. Determine the resource profile for this project if the activities are to start on their

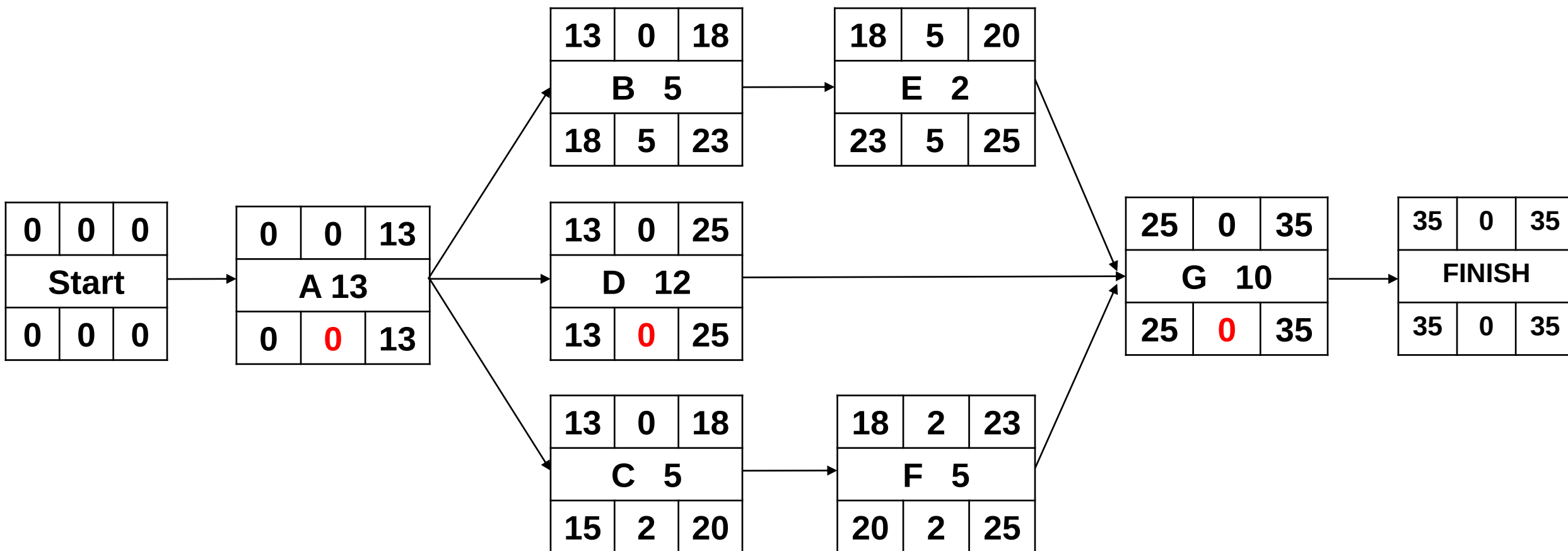
- i) Earliest start time
- ii) Latest start time

Using the floats of activities, obtain an optimized resource profile.

Activity	Predecessors	Duration (days)	No. of resources
A	--	13	2
B	A	5	3
C	A	5	4
D	A	12	2
E	B	2	3
F	C	5	3
G	D, E, F	10	4

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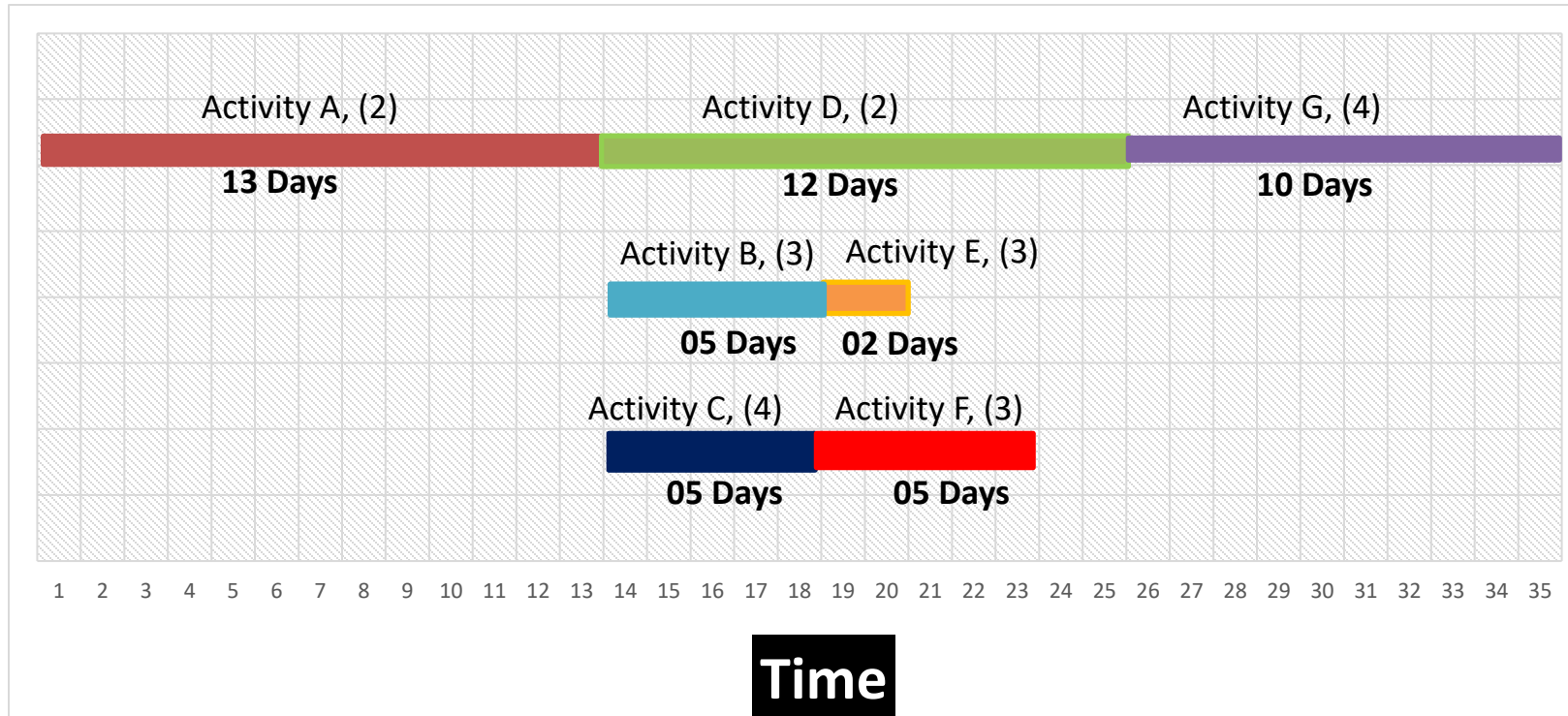


Activities **A**, **D** and **G** are critical, and the project duration is **35 days**

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Arranging the activities in the order of their earliest start times



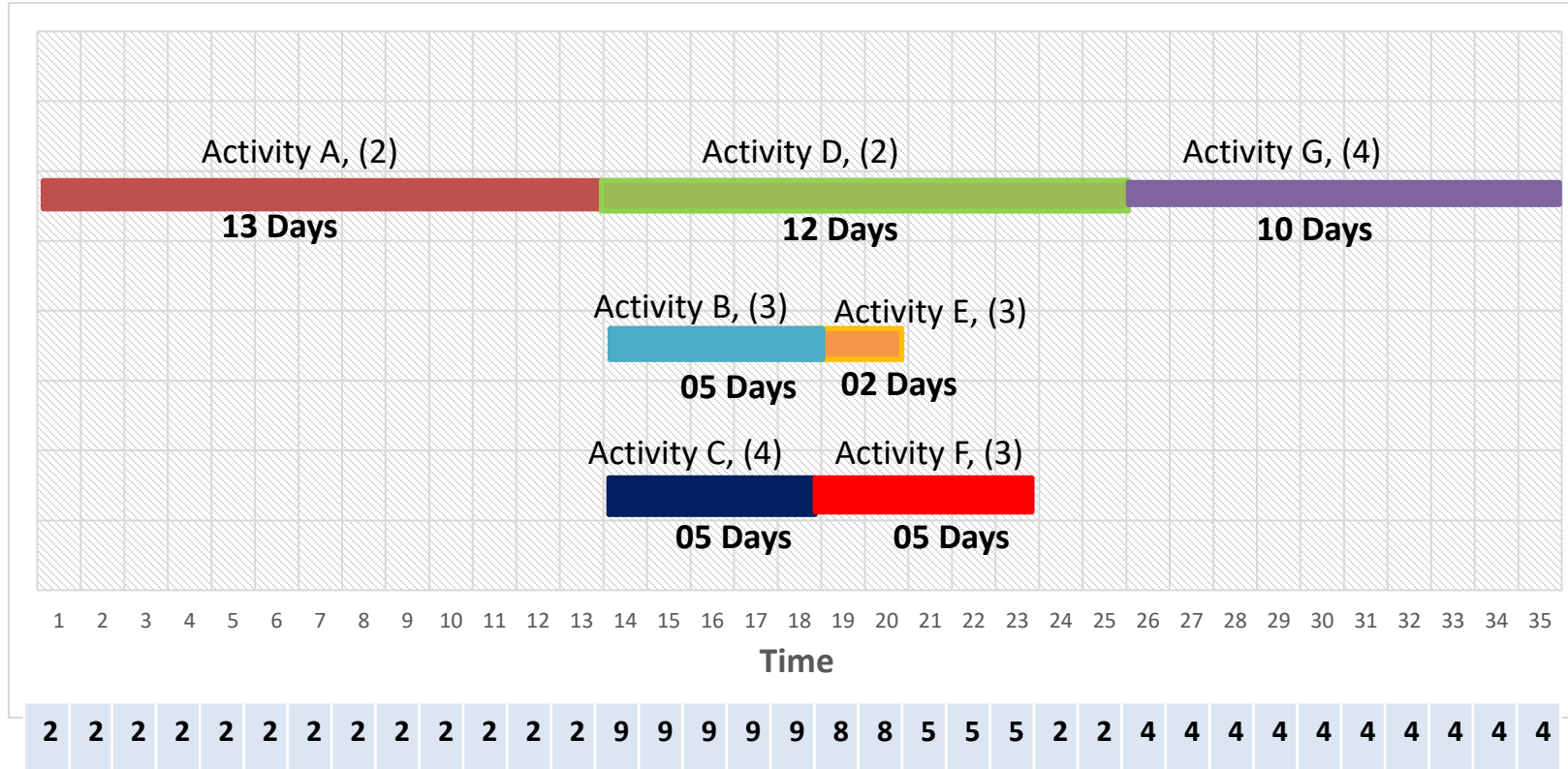
Critical activities are represented on the top

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Arranging the activities in the order of their earliest start times

Resource
loading based
on EST

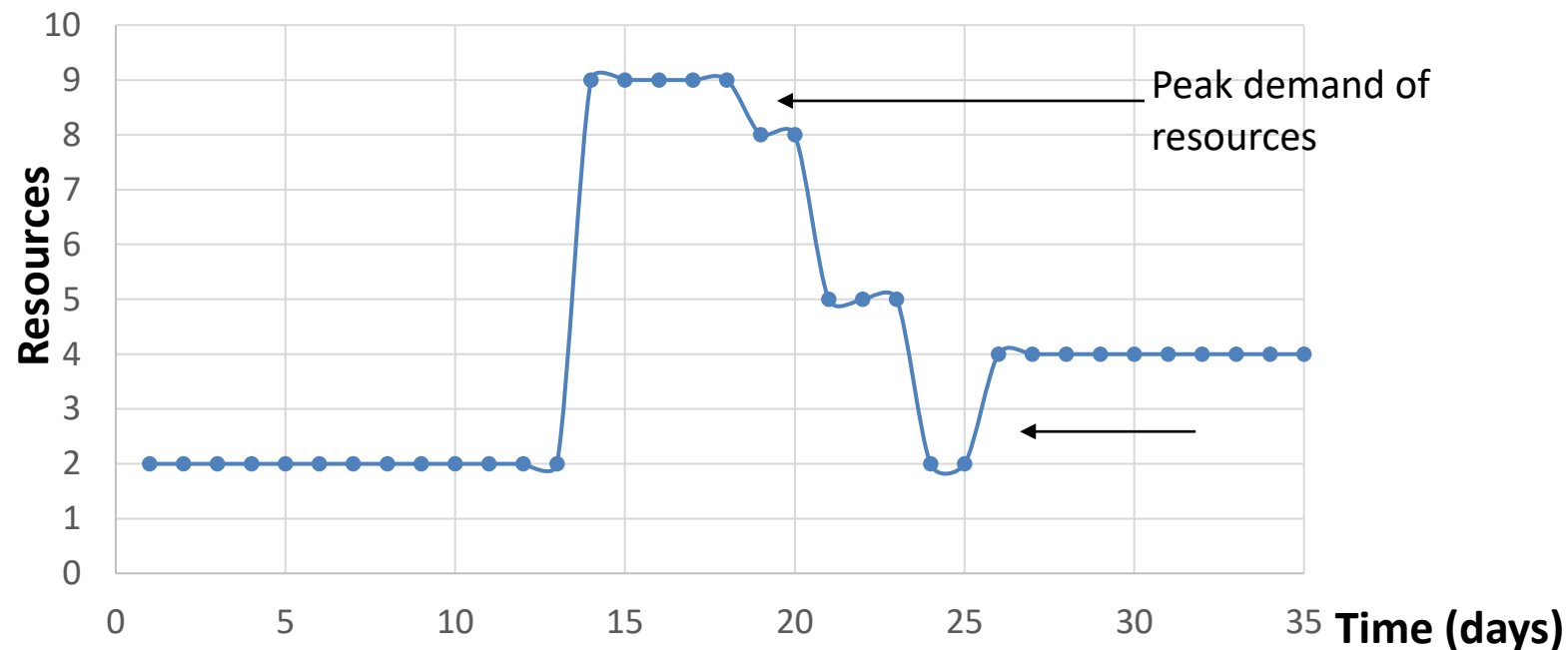


Critical activities are represented on the top

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Graphical representation of resource profile, with activities starting at EST

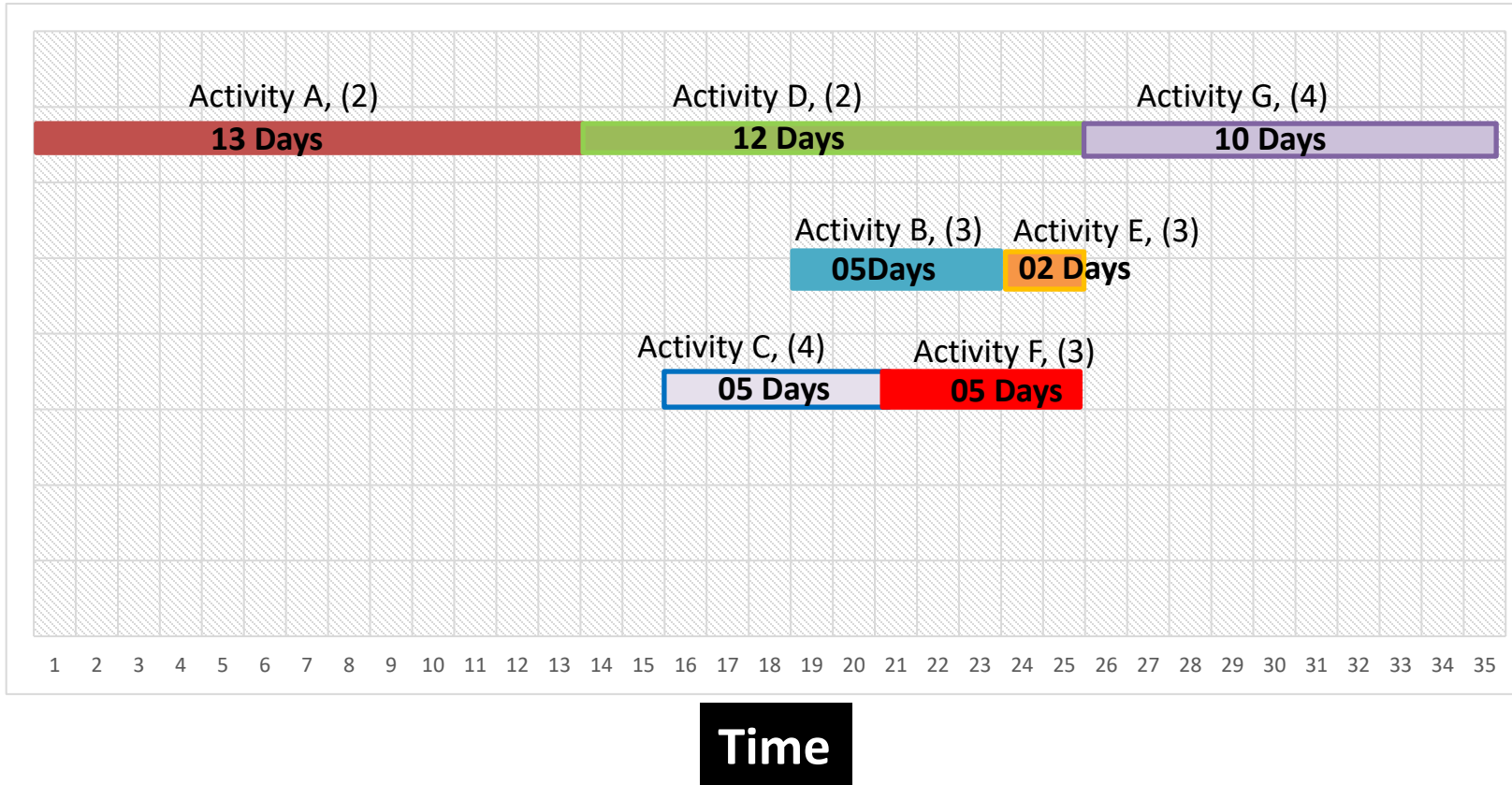


Note: There is a sudden rise in the demand of resources in the middle of the project.

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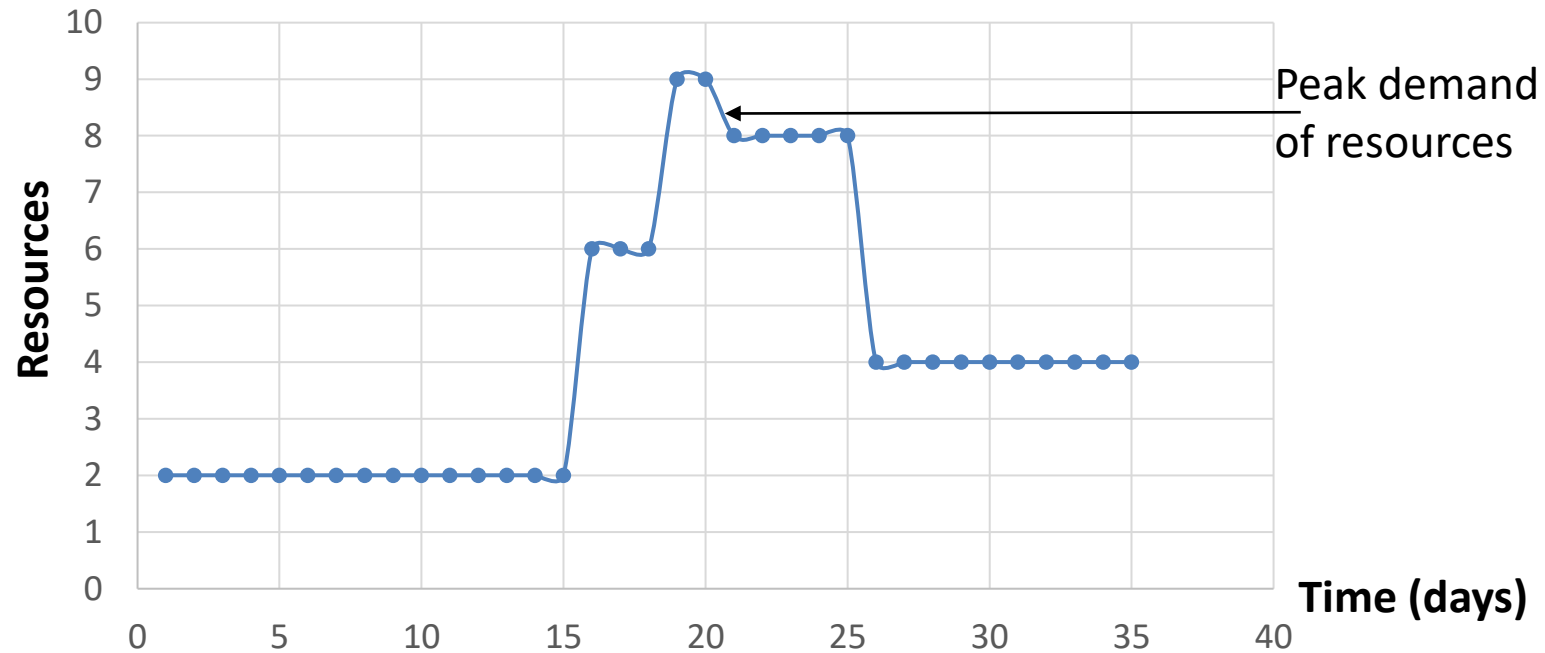
Arranging the activities in the order of their latest start times



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Graphical representation of resource profile, with activities starting on LST



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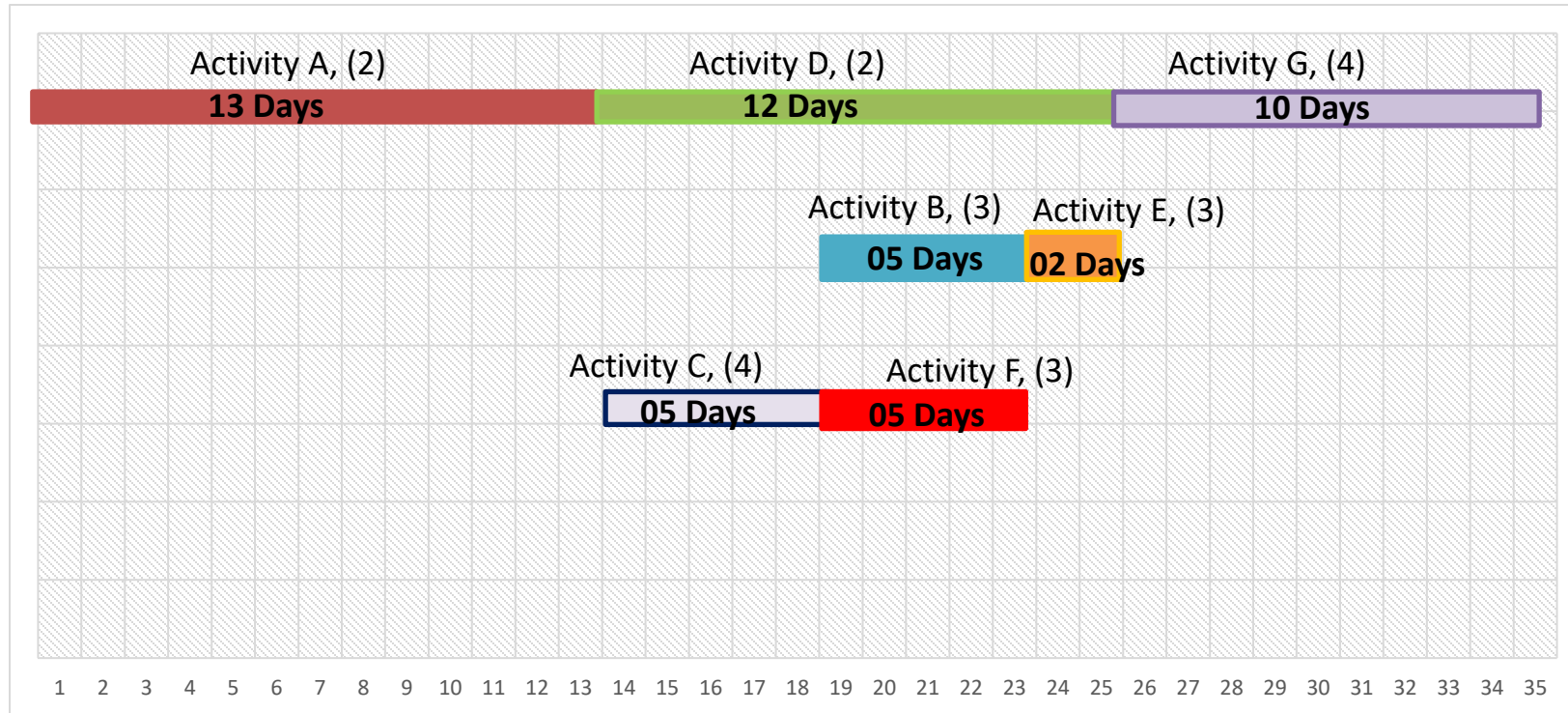
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In this example, activities 'B' and 'E' have a float of 5 days, whereas both 'C' and 'F' have a float of 2 days. This can be kept in mind and activities can be scheduled accordingly. It may help in obtaining gradual change in resource deployment, and it may help in reducing the peak demand also.

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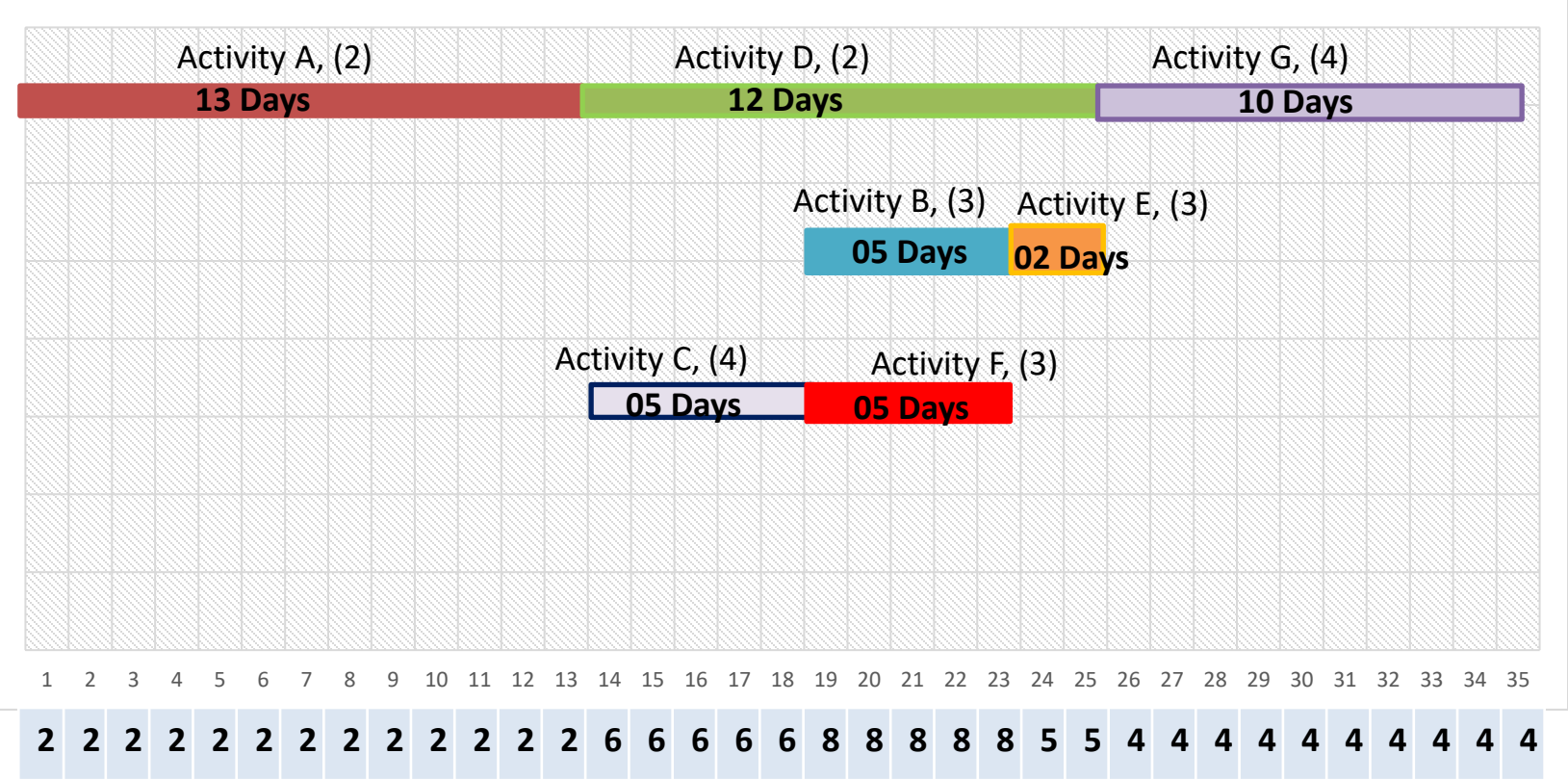
Let activities C and F are starting on their EST, and, activities B and E are starting on their LST.



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Let activities C and F are starting on their EST, and, activities B and E are starting on their LST.

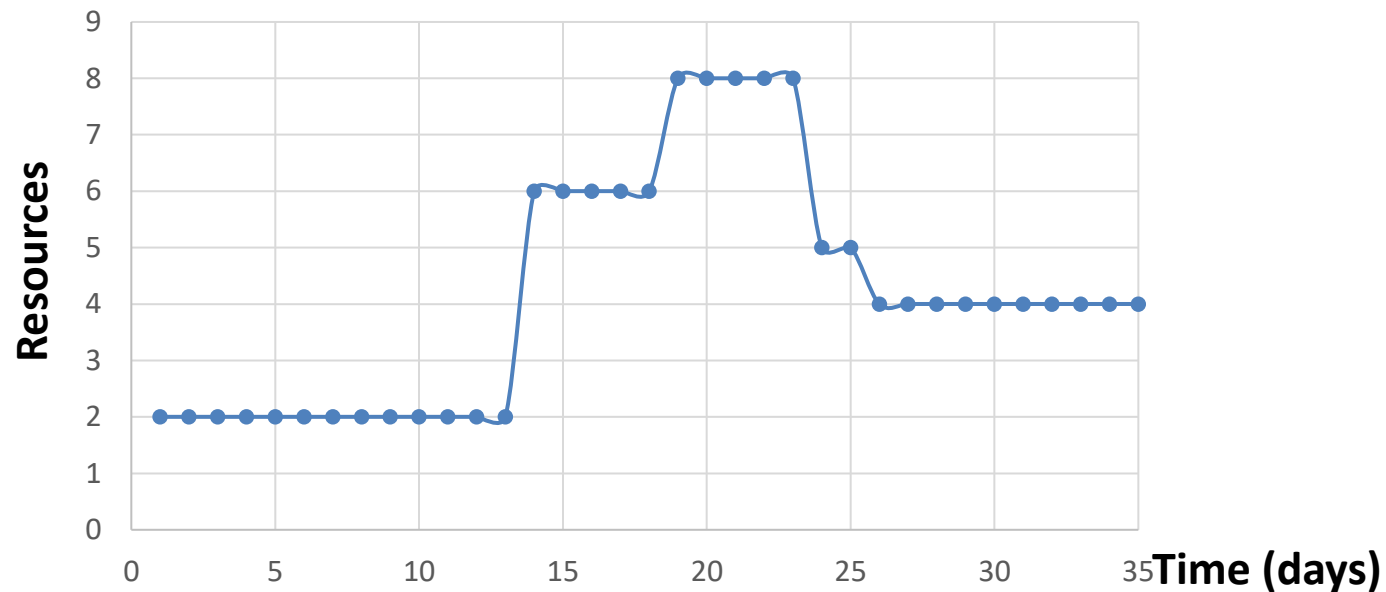


Resource loading

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Graphical representation of resource profile

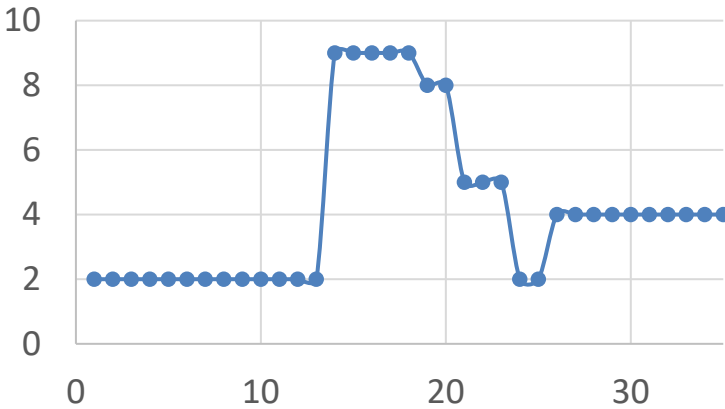


The peak demand of resources has reduced from 9 to 8 and the profile has gradual changes in the resource requirement, which is more desirable

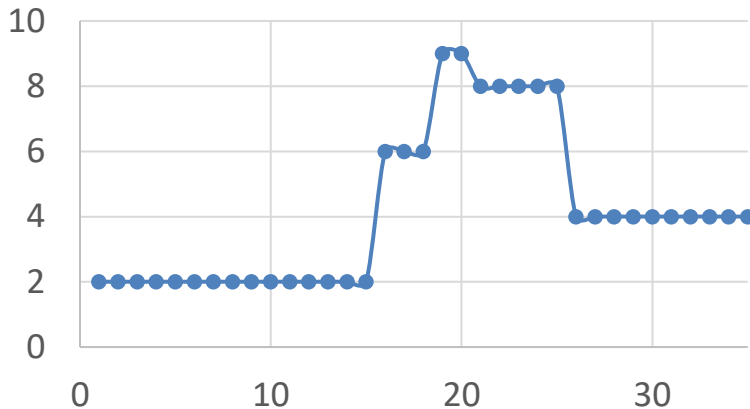
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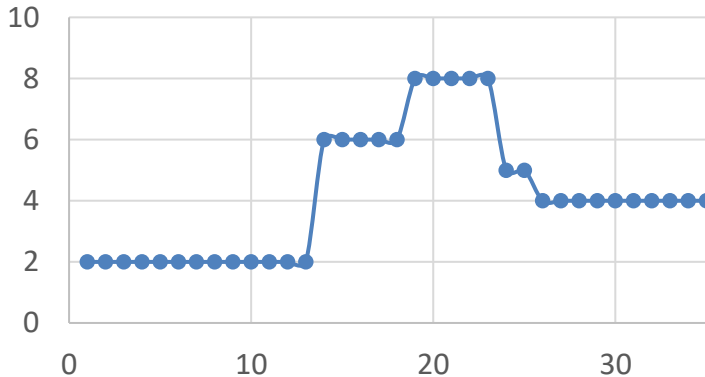
Comparative Analysis



Resource distribution based on
EST



Resource distribution based on
LST



Resource distribution based on combination of
EST and LST

RESOURCE LEVELING

- Resource leveling is performed when the availability of the resources is a constraint.
- Due to limited availability of resources, some activities may have to be delayed.
- The basic objective is to find out which activities can be delayed and finally arrive at the shortest possible time to complete the project satisfying the resource constraints.
- This may or may not result in the delay of the project.

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Resource leveling can be carried out through the following steps:

1. Identify the critical activities in the network
2. Allocate the resources day-wise
3. If at any time, the resource requirement exceeds availability, then allocation of resources to activities should follow the rules below
 - Rule-1:- Critical activities get the first preference
 - Rule-2:- On-going activities should not be stopped in the middle
 - Rule-3:- Allocate resources to an activity that has the least EST
 - Rule-4:- If two activities have same EST, then allocate resources to the activity that has lesser float

Notes:

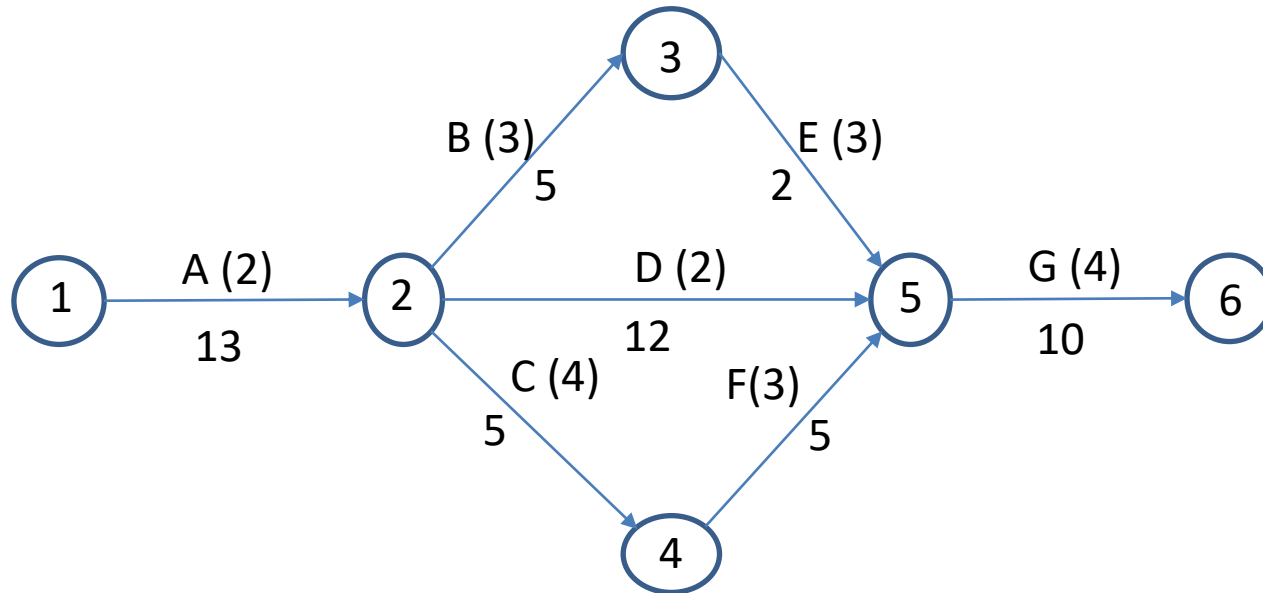
1. The activities may be rescheduled due to resource leveling, which may result in changes in activities that are critical.
2. It should be ensured that the above is accounted for in making a decision based on the rules.

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ILLUSTRATIVE EXAMPLE

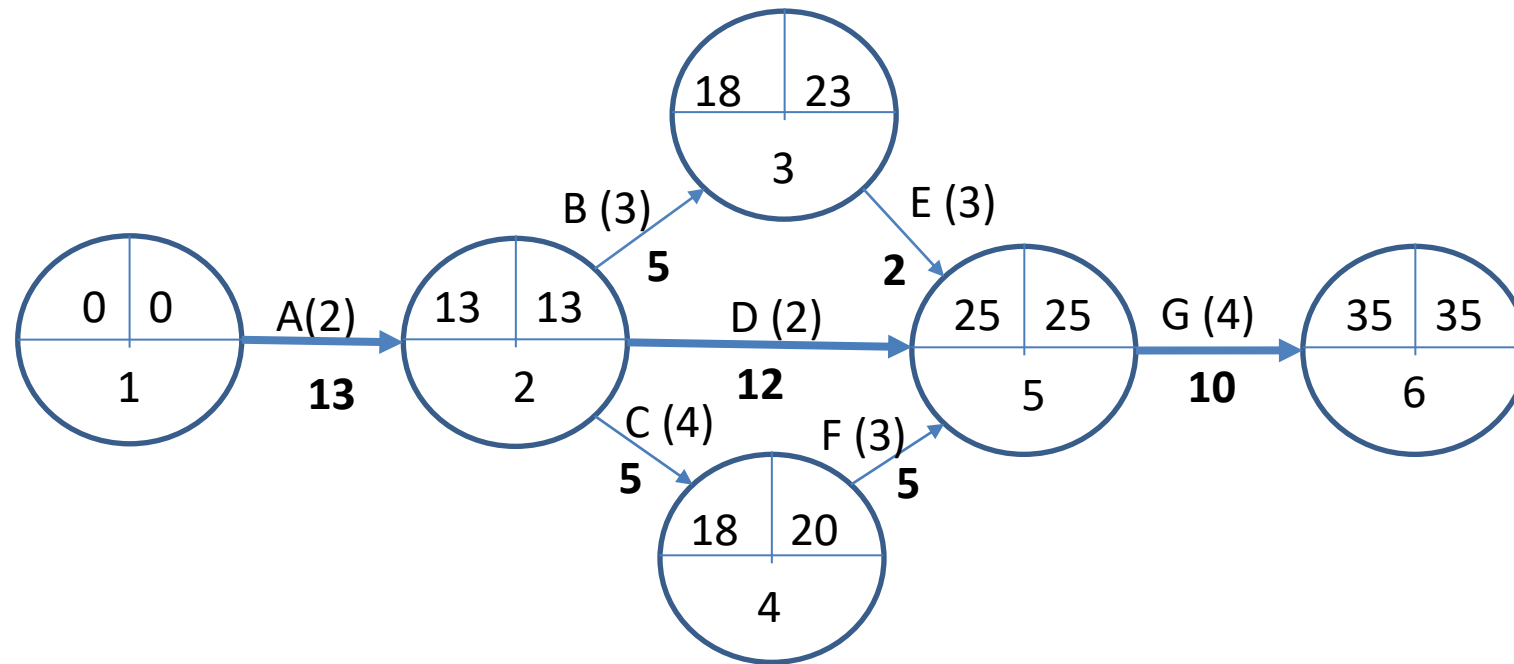
Let us consider the same example in which the project takes 35 days to complete normally. Assume that the maximum availability of resources on a given day is 8. Determine if there is any delay in the project completion.



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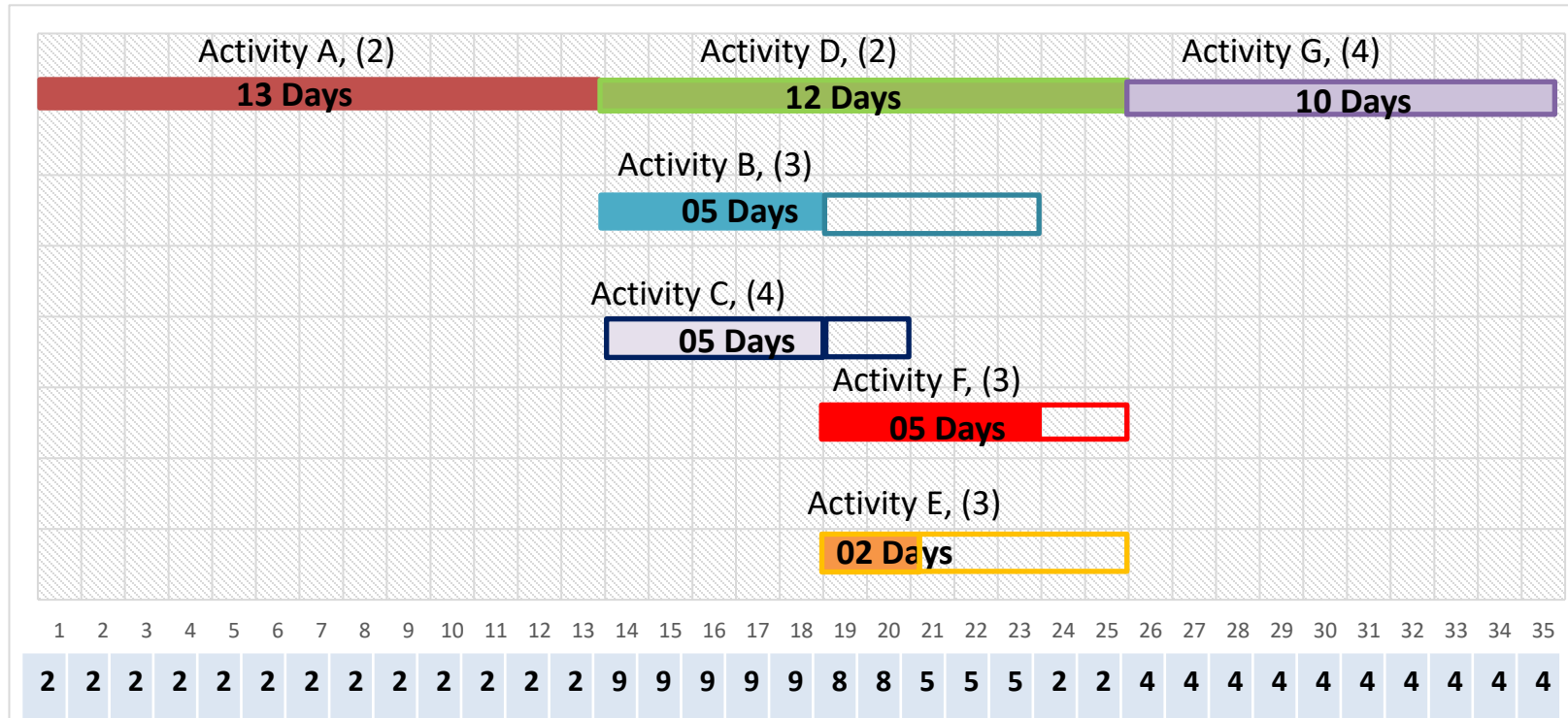
Earliest and latest occurrences of events are shown on nodes
(obtained by carrying out forward pass and backward pass)



Activities A, D and G are critical, and the project duration is 35 days

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Graphical representation of activities starting at EST (with their float)



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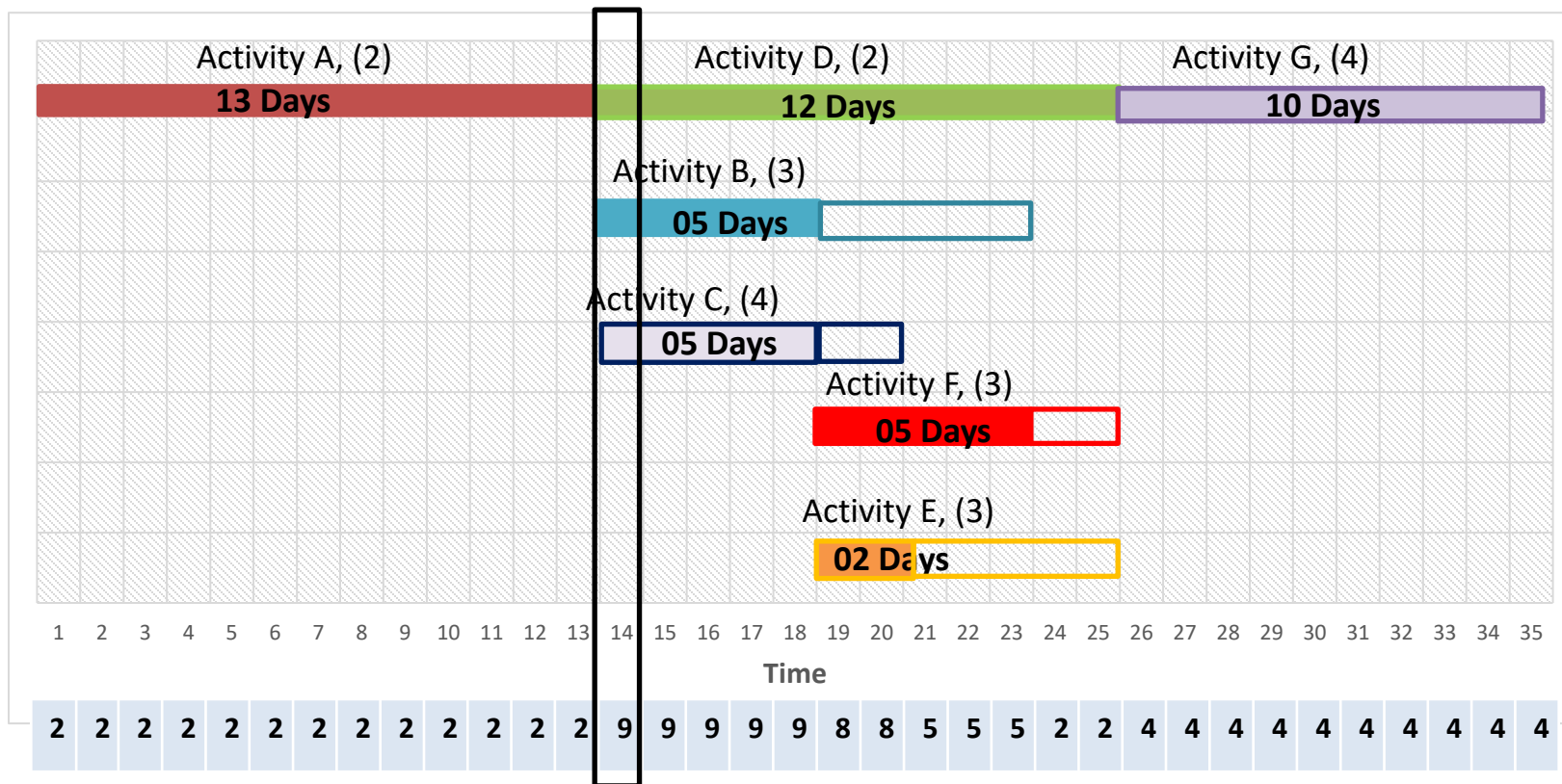
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Till 13 days, activity A having a resource requirement of 2 is carried out.

On Day-14, resources need to be allocated to activities 'D', 'B' and 'C', if they have to start on their respective EST.

The total resource requirement is 9, which is exceeding the available resource limit.

Therefore, priority rules should be used to decide the allocation.



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Day 14

Activities	Resource requirement	Critical (Y/N)	On-Going	Earliest Start Time	Float	Priority
D	2	Y	N	13	0	1st
B	3	N	N	13	5	3rd
C	4	N	N	13	2	2nd

Thus, activities 'D' and 'C' will be scheduled together constituting SIX resources.

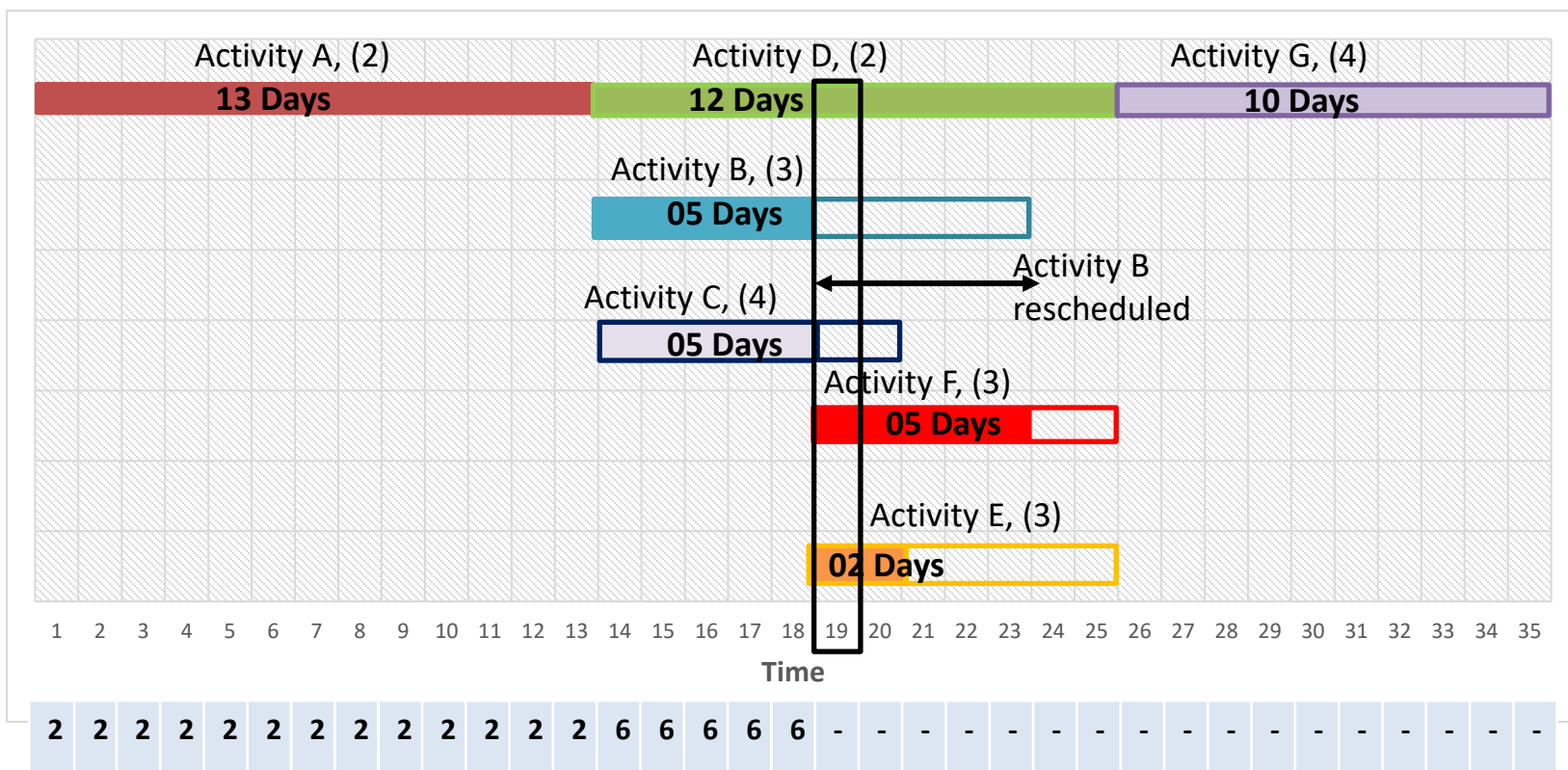
And, activity 'B' will be delayed.

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Since resources cannot be allocated to B till C finishes, the same resource allocation will be followed till the end of 18th day.

Therefore, on Day-19, Activities 'D','B','E' and 'F' are competing for the resources.



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Day 19

Activities	Resource requirement	Critical Activity (Y/N)	On-Going	Earliest Start Time	Float	Priority
D	2	Y	Y	13	0	1st
B	3	N	N	13	5	2nd
E	3	N	N	18	5	4th
F	3	N	N	18	2	3rd

Therefore, activities 'D' , 'B' and 'F' will be started (total resource requirement is 8).

And, activity 'E' is delayed.

Note that Activity 'B' has delayed by 5 days from its EST.

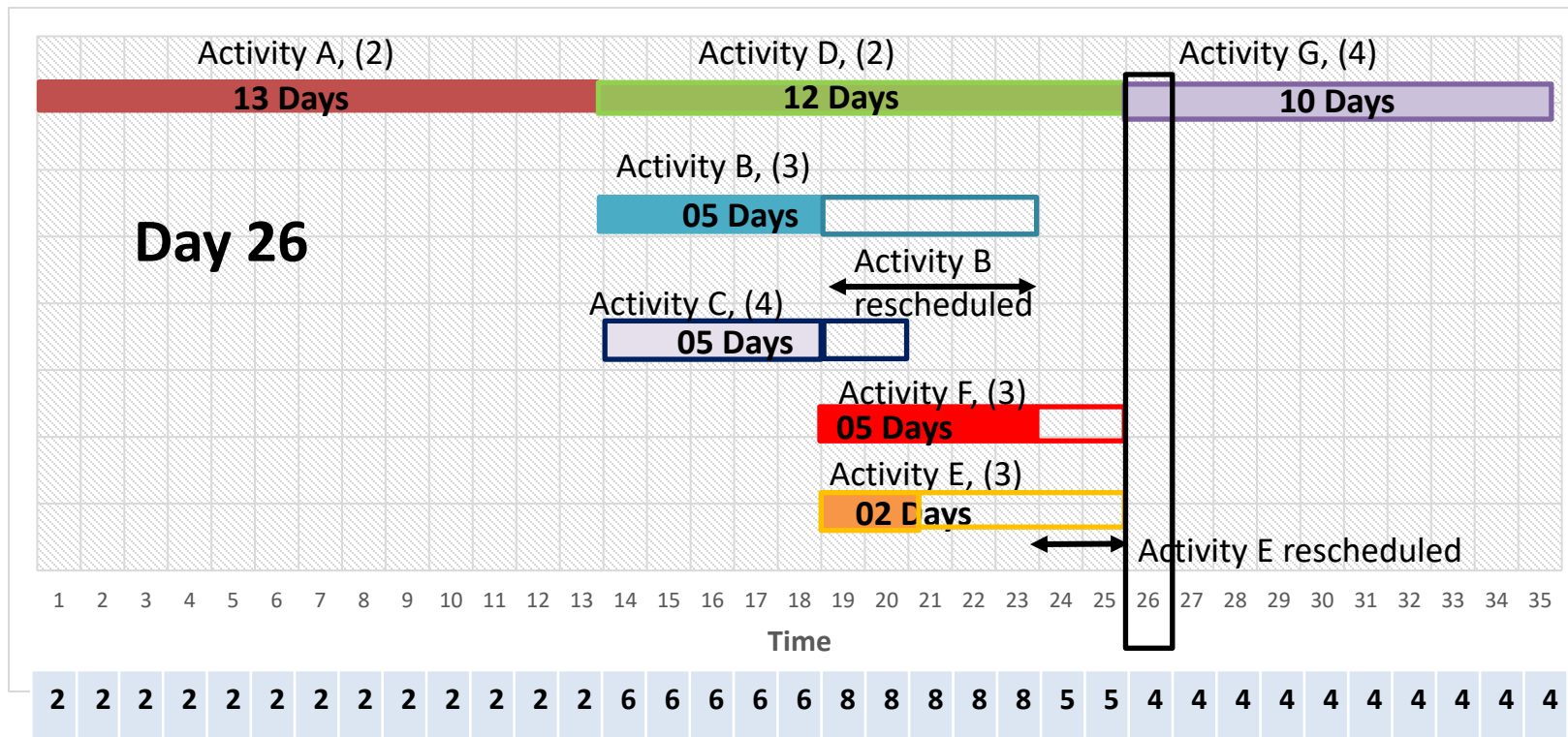
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- All the preceding activities A, B, C, D, E and F would be completed by Day 25.
- So resources will be allocated to critical activity 'G' which has resource requirement of 4.

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Resource loading based on revised schedule of activities

Activity 'G' shall be allocated with required resources of 4 and the project shall be completed within their scheduled time of 35 days.

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- There is a delay in the start time of activities (B and E), due to resource limitation of 8.
- But these activities are completed within their float time.
- Therefore, the project has not been delayed at the end.

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Thank you